

Kubernetes Assignment Solution

1. Install minikube on a machine.

```
anilavaghosh@Amitavas-MBP addressbook-demo % curl -LO https://storage.googleapis.com/minikube/releases/latest/minikube-darwin-arm64
sudo install minikube-darwin-arm64 /usr/local/bin/minikube

% Total % Received % Xferd Average Speed Time Time Time Current
100 87.5M 100 87.5M 0 0 51.5M 0 0:00:01 0:00:01 --:--:-- 51.5M
Password:
anilavaghosh@Amitavas-MBP addressbook-demo %
```

2. Start minikube cluster.

minikube start

Creating alias for **minikube kubectl** —

alias kubectl="minikube kubectl --"
kubectl get pods -A

```

anitavaghosh@Mitavias-MBP addressbook-demo % curl -LO https://storage.googleapis.com/minikube/releases/latest/minikube-darwin-arm64
sudo install minikube-darwin-arm64 /usr/local/bin/minikube

% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
100 87.0M 100 87.0M 0 0 51.0M 0 0:00:01 0:00:01 --|-- 51.0M

Password:
anitavaghosh@Mitavias-MBP addressbook-demo % minikube start
minikube v1.32.0 on Darwin 14.2.1 (arm64)
* Automatically selected the docker driver
* Using Docker Desktop driver with root privileges
* Starting control plane node minikube in cluster minikube
* Pulling base image ...
* Downloading Kubernetes v1.28.3 preload ...
> preloaded-images-k8s-v18-v1...: 341.16 MiB / 341.16 MiB 100.00% 61.07 M
* Creating Docker container (CPUs=2, Memory=4096MB) ...
* Preparing Kubernetes v1.28.3 on Docker 24.0.7 ...
* Generating certificates and keys ...
* Booting up control plane ...
* Configuring RBAC rules ...
* Configuring bridge CNI (Container Networking Interface) ...
* Verifying Kubernetes components...
* Using image gcr.io/k8s-minikube/storage-provisioner:v5
* Enabled addons: storage-provisioner, default-storageclass
* kubectl not found, if you need it, try: 'minikube kubectl -- get pods -A'
* Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
anitavaghosh@Mitavias-MBP addressbook-demo % kubectl get services
> kubectl get services
> kubectl get services
> kubectl get services
NAME TYPE CLUSTER-IP EXTERNAL-IP PORT(S) AGE
minikube ClusterIP 10.96.0.1 <none> 443/TCP 57s
anitavaghosh@Mitavias-MBP addressbook-demo % kubectl get pods -A
NAME NAMESPACE NAME READY STATUS RESTARTS AGE
minikube kube-system coredns-5dd575bbd8-c4ppg 1/1 Running 0 71s
minikube kube-system etcd-minikube 1/1 Running 0 88s
minikube kube-system kube-apiserver-minikube 1/1 Running 0 88s
minikube kube-system kube-controller-manager-minikube 1/1 Running 0 88s
minikube kube-system kube-proxy-rzk22 1/1 Running 0 71s
minikube kube-system kube-scheduler-minikube 1/1 Running 0 88s
minikube kube-system storage-provisioner 1/1 Running 1 (67s ago) 84s
anitavaghosh@Mitavias-MBP addressbook-demo %

```

3. Write a deployment.yml file to deploy addressbook application.

deployment.yml

=====

apiVersion: apps/v1

kind: Deployment

metadata:

name: addressbook

labels:

app: addressbook

spec:

replicas: 1

selector:

matchLabels:

app: addressbook

template:

metadata:

labels:

app: addressbook

spec:
 containers:
 - name: addressbook
 image: amitspring/addressbook-demo:3.0
 ports:
 - containerPort: 80

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: addressbook
  labels:
    app: addressbook
spec:
  replicas: 1
  selector:
    matchLabels:
      app: addressbook
  template:
    metadata:
      labels:
        app: addressbook
    spec:
      containers:
        - name: addressbook
          image: amitspring/addressbook:3.0
          ports:
            - containerPort: 80
```

4. Create the deployment with deployment.yml.

kubectl create -f Deployment.yml

```

antavaghosh@Mintavias-MBP addressbook-demo % curl -LO https://storage.googleapis.com/minikube/releases/latest/minikube-darwin-arm64
sudo install minikube-darwin-arm64 /usr/local/bin/minikube

% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
100 87.0M 100 87.0M 0 0 51.0M 0 0:00:01 0:00:01 --:--:-- 51.0M

Password:
antavaghosh@Mintavias-MBP addressbook-demo % minikube start
minikube v1.32.0 on Darwin 14.2.1 (arm64)
* Automatically selected the docker driver
* Using Docker Desktop driver with root privileges
* Starting control plane node minikube in cluster minikube
* Pulling base image ...
* Downloading Kubernetes v1.28.3 preload ...
> preloaded-images-k8s-v18-v1...: 341.16 MiB / 341.16 MiB 100.00% 61.07 M
* Creating Docker container (CPUs=2, Memory=4800M) ...
* Preparing Kubernetes v1.28.3 on Docker 24.0.7 ...
* Generating certificates and keys ...
* Booting up control plane ...
* Configuring RBAC rules ...
* Configuring bridge CNI (Container Networking Interface) ...
* Verifying Kubernetes components...
* Using image gcr.io/k8s-minikube/storage-provisioner:v5
Enabled addons: storage-provisioner, default-storageclass
Kubectl not found. If you need it, try: 'minikube kubectl -- get pods -A'
Don't kubectl is now configured to use "minikube" cluster and "default" namespace by default
antavaghosh@Mintavias-MBP addressbook-demo % alias kubectl="minikube kubectl --"
antavaghosh@Mintavias-MBP addressbook-demo % kubectl get services
> kubectl.sh:254: 64 B / 64 B [-----] 100.00% ? p/s 0s
> kubectl: 52.80 MiB / 52.80 MiB [-----] 100.00% 5.01 MiB p/s 11s

NAME      TYPE      CLUSTER-IP   EXTERNAL-IP   PORT(S)    AGE
kubernetes ClusterIP   10.96.0.1     <none>         443/TCP     57s

antavaghosh@Mintavias-MBP addressbook-demo % kubectl get pods -A
NAMESPACE   NAME      READY   STATUS    RESTARTS   AGE
kube-system   coredns-5d575bb8-4qpg  1/1     Running   0           71s
kube-system   etcd-minikube  1/1     Running   0           86s
kube-system   kube-apiserver-minikube  1/1     Running   0           86s
kube-system   kube-controller-manager-minikube  1/1     Running   0           86s
kube-system   kube-proxy-rzk22  1/1     Running   0           71s
kube-system   kube-scheduler-minikube  1/1     Running   0           86s
kube-system   storage-provisioner  1/1     Running   1 (67s ago)  84s

antavaghosh@Mintavias-MBP addressbook-demo % vi Deployment.yaml
antavaghosh@Mintavias-MBP addressbook-demo % kubectl create -f Deployment.yaml
deployment.apps/addressbook created
antavaghosh@Mintavias-MBP addressbook-demo % kubectl get pods
NAME      READY   STATUS    RESTARTS   AGE
addressbook-5d86d576f9-zhm5  0/1     ContainerCreating   0           6s

antavaghosh@Mintavias-MBP addressbook-demo % kubectl get pods
NAME      READY   STATUS    RESTARTS   AGE
addressbook-5d86d576f9-zhm5  1/1     Running            0           12s
antavaghosh@Mintavias-MBP addressbook-demo %

```

5. Scale up the addressbook application up to 3 instances.

```

apiVersion: apps/v1
kind: Deployment
metadata:
  name: addressbook
  labels:
    app: addressbook
spec:
  replicas: 3
  selector:
    matchLabels:
      app: addressbook
  template:
    metadata:
      labels:
        app: addressbook
    spec:
      containers:
        - name: addressbook
          image: amitspring/addressbook:3.0
          ports:
            - containerPort: 80

```



```

antavaghash@Antivas-MBP addressbook-demo % curl -LO https://storage.googleapis.com/minikube/releases/latest/minikube-darwin-arm64
sudo install minikube-darwin-arm64 /usr/local/bin/minikube

% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
100 87.0M 100 87.0M 0 0 51.0M 0 0:00:01 0:00:01 --|-- 51.0M

Password:
antavaghash@Antivas-MBP addressbook-demo % minikube start
minikube v1.32.0 on Darwin 14.2.1 (arm64)
Automatically selected the docker driver
Using Docker Desktop driver with root privileges
Starting control plane node minikube in cluster minikube
Pulling base image ...
Downloading Kubernetes v1.28.3 preload ...
> preloaded-images-k8s-v18-v1...: 341.16 MiB / 341.16 MiB 100.00% 61.07 M
Creating Docker container (CPUs=2, Memory=4096M) ...
Preparing Kubernetes v1.28.3 on Docker 24.0.7 ...
  * Generating certificates and keys ...
  * Booting up control plane ...
  * Configuring RBAC rules ...
  * Configuring bridge CNI (Container Networking Interface) ...
  * Verifying Kubernetes components...
  * Using image gcr.io/k8s-minikube/storage-provisioner:v5
  * Enabled addons: storage-provisioner, default-storageclass
kubectctl not found. If you need it, try: 'minikube kubectctl --get pods -A'
Don't kubectctl is now configured to use "minikube" cluster and "default" namespace by default
antavaghash@Antivas-MBP addressbook-demo % alias kubectctl="minikube kubectctl --"
antavaghash@Antivas-MBP addressbook-demo % kubectctl get services
> kubectctl.sh25d: 64.8 / 64.8 [-----] 100.00% 7 p/s 0s
> kubectctl: 52.80 MiB / 52.80 MiB [-----] 100.00% 5.01 MiB p/s 11s

NAME                TYPE                CLUSTER-IP    EXTERNAL-IP    PORT(S)    AGE
kubernetes           ClusterIP           10.96.0.1      <none>          443/TCP     57s
antavaghash@Antivas-MBP addressbook-demo % kubectctl get pods -A
NAMESPACE   NAME                READY   STATUS    RESTARTS   AGE
kube-system  coredns-5d575abb8-c4pqa  1/1     Running   0           71s
kube-system  etcd-minikube         1/1     Running   0           85s
kube-system  kube-apiserver-minikube  1/1     Running   0           85s
kube-system  kube-controller-manager-minikube  1/1     Running   0           85s
kube-system  kube-proxy-rzk22       1/1     Running   0           71s
kube-system  kube-scheduler-minikube  1/1     Running   0           85s
kube-system  storage-provisioner     1/1     Running   1 (67s ago)  84s
antavaghash@Antivas-MBP addressbook-demo % vi Deployment.yaml
antavaghash@Antivas-MBP addressbook-demo % kubectctl create -f Deployment.yaml
deployment.apps/addressbook created
antavaghash@Antivas-MBP addressbook-demo % kubectctl get pods
NAME                READY   STATUS    RESTARTS   AGE
addressbook-5686d576f9-zhm5  0/1     ContainerCreating   0           6s
antavaghash@Antivas-MBP addressbook-demo % kubectctl get pods
NAME                READY   STATUS    RESTARTS   AGE
addressbook-5686d576f9-zhm5  1/1     Running      0          12s
antavaghash@Antivas-MBP addressbook-demo % vi Deployment.yaml
antavaghash@Antivas-MBP addressbook-demo % kubectctl apply -f Deployment.yaml
Warning: resource Deployment/addressbook is missing the kubectctl.kubernetes.io/last-applied-configuration annotation which is required by kubectctl apply. kubectctl apply should only be used on resources created declaratively by either kubectctl create --save-conf or kubectctl apply. The missing annotation will be patched automatically.
deployment.apps/addressbook configured
antavaghash@Antivas-MBP addressbook-demo % kubectctl get pods
NAME                READY   STATUS    RESTARTS   AGE
addressbook-5686d576f9-j84k  1/1     Running      0           6s
addressbook-5686d576f9-zhm5  1/1     Running      0          2022s
addressbook-5686d576f9-x6gs2  1/1     Running      0           6s
antavaghash@Antivas-MBP addressbook-demo %

```

6. Scale down the addressbook application to 1 instance.

```

apiVersion: apps/v1
kind: Deployment
metadata:
  name: addressbook
  labels:
    app: addressbook
spec:
  replicas: 1
  selector:
    matchLabels:
      app: addressbook
  template:
    metadata:
      labels:
        app: addressbook
    spec:
      containers:
        - name: addressbook
          image: smitspring/addressbook:3.0
          ports:
            - containerPort: 80

```

```

100 87.5M 100 87.5M 0 0 51.5M 0 0:00:01 0:00:01 --|-- 51.5M
Password:
minikube v1.32.0 on Darwin 14.2.1 (arm64)
Automatically selected the docker driver
Using Docker Desktop driver with root privileges
Starting control plane node minikube in cluster minikube
Pulling base image ...
Downloading Kubernetes v1.28.3 preload ...
> preload-images-v1.28.3-... 341.1s MiB / 341.1s MiB 100.00% 61.07 M
Creating docker container (CPUs=2, Memory=4096MB) ...
Preparing Kubernetes v1.28.3 on Docker 24.0.7 ...
  * Generating certificates and keys ...
  * Booting up control plane ...
  * Configuring RBAC rules ...
  * Configuring bridge CNI (Container Networking Interface) ...
  * Verifying Kubernetes components...
  * Using image gcr.io/k8s-minikube/storage-provisioner:v5
  * Enabled addons: storage-provisioner, default-storageclass
Kubectl not found. If you need it, try: 'minikube kubectl -- get pods -A'
Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
minikube kubectl -- addressbook-demo % alias kubectl=minikube kubectl --
minikube kubectl -- addressbook-demo % kubectl get services
> kubectl -naddressbook get pods
NAME                                READY    STATUS    RESTARTS   AGE
addressbook-5686d576f9-zhnm5        0/1      ContainerCreating    0          6s
minikube kubectl -- addressbook-demo % kubectl get pods
NAME                                READY    STATUS    RESTARTS   AGE
addressbook-5686d576f9-zhnm5        1/1      Running    0          12s
minikube kubectl -- addressbook-demo % kubectl apply -f Deployment.yaml
Warning: resource deployments/addressbook is missing the kubectll.kubernetes.io/last-applied-configuration annotation which is required by kubectl apply. kubectl apply should only be used on resources created declaratively by either kubectl create --save-config or kubectl apply. The missing annotation will be patched automatically.
minikube kubectl -- addressbook-demo % kubectl get pods
NAME                                READY    STATUS    RESTARTS   AGE
addressbook-5686d576f9-j854k        1/1      Running    0          6s
addressbook-5686d576f9-zhnm5        1/1      Running    0          2022s
addressbook-5686d576f9-xgqz2        1/1      Running    0          4s
minikube kubectl -- addressbook-demo % kubectl apply -f Deployment.yaml
minikube kubectl -- addressbook-demo % kubectl get pods
NAME                                READY    STATUS    RESTARTS   AGE
addressbook-5686d576f9-j854k        1/1      Running    0          72s
addressbook-5686d576f9-zhnm5        1/1      Running    0          3m48s
addressbook-5686d576f9-xgqz2        1/1      Running    0          92s
minikube kubectl -- addressbook-demo % kubectl apply -f Deployment.yaml
minikube kubectl -- addressbook-demo % kubectl get pods
NAME                                READY    STATUS    RESTARTS   AGE
addressbook-5686d576f9-zhnm5        1/1      Running    0          3m58s
minikube kubectl -- addressbook-demo %

```

7. expose your addressbook application deployment using a Kubernetes service.

kubectl expose deployment addressbook --type=NodePort --port=8080

```

Starting control plane node minikube in cluster minikube
Pulling base image ...
Downloading Kubernetes v1.28.3 preload ...
> preloaded-images-k8s-v18-v1...: 341.16 MiB / 341.16 MiB 100.00% 61.07 M
Creating docker container (CPUs=2, Memory=4096MB) ...
Preparing Kubernetes v1.28.3 on Docker 24.0.7 ...
  • Generating certificates and keys ...
  • Booting up control plane ...
  • Configuring RBAC rules ...
  • Configuring bridge CNI (Container Networking Interface) ...
  • Verifying Kubernetes components...
  • Using image gcr.io/k8s-minikube/storage-provisioner:v5
  Enabled add-ons: storage-provisioner, default-storageclass
Kubectl not found. If you need it, try: 'minikube kubectl -- get pods -A'
Done! Kubectl is now configured to use "minikube" cluster and "default" namespace by default
antivaghash@Antivas-MBP addressbook-demo % kubectl get services
antivaghash@Antivas-MBP addressbook-demo % kubectl get pods
> kubectl get pods
> kubectl: 64.8 / 64.8 MiB [-----] 100.00% 0.01 MiB p/s 11s
NAME                TYPE                CLUSTER-IP      EXTERNAL-IP      PORT(S)          AGE
kubernetes           ClusterIP           10.96.0.1        <none>            443/TCP          57s
antivaghash@Antivas-MBP addressbook-demo % kubectl get pods -A
NAMESPACE   NAME                 READY   STATUS    RESTARTS   AGE
kube-system  coredns-5dd575ab8-c4pgn  1/1     Running   0           71s
kube-system  etcd-minikube         1/1     Running   0           88s
kube-system  kube-apiserver-minikube  1/1     Running   0           88s
kube-system  kube-controller-manager-minikube  1/1     Running   0           88s
kube-system  kube-proxy-rzk22      1/1     Running   0           71s
kube-system  kube-scheduler-minikube  1/1     Running   0           88s
kube-system  storage-provisioner    1/1     Running   1 (67s ago)  84s
antivaghash@Antivas-MBP addressbook-demo % vi Deployment.yaml
antivaghash@Antivas-MBP addressbook-demo % kubectl create -f Deployment.yaml
deployment.apps/addressbook created
antivaghash@Antivas-MBP addressbook-demo % kubectl get pods
NAME                READY   STATUS    RESTARTS   AGE
addressbook-5686d576f9-zhm5  0/1     ContainerCreating   0           6s
antivaghash@Antivas-MBP addressbook-demo % kubectl get pods
NAME                READY   STATUS    RESTARTS   AGE
addressbook-5686d576f9-zhm5  1/1     Running        0          12s
antivaghash@Antivas-MBP addressbook-demo % vi Deployment.yaml
antivaghash@Antivas-MBP addressbook-demo % kubectl apply -f Deployment.yaml
Warning: resource Deployment/addressbook is missing the kubectrl.kubernetes.io/last-applied-configuration annotation which is required by kubectl apply. kubectl apply should only be used on resources created directly by either kubectl create --save-config or kubectl apply. The missing annotation will be patched automatically.
deployment.apps/addressbook configured
antivaghash@Antivas-MBP addressbook-demo % kubectl get pods
NAME                READY   STATUS    RESTARTS   AGE
addressbook-5686d576f9-j854k  1/1     Running        0           6s
addressbook-5686d576f9-zhm5  1/1     Running        0          2022s
addressbook-5686d576f9-xgq2  1/1     Running        0           6s
antivaghash@Antivas-MBP addressbook-demo % vi Deployment.yaml
antivaghash@Antivas-MBP addressbook-demo % kubectl get pods
NAME                READY   STATUS    RESTARTS   AGE
addressbook-5686d576f9-j854k  1/1     Running        0           92s
addressbook-5686d576f9-zhm5  1/1     Running        0          3644s
addressbook-5686d576f9-xgq2  1/1     Running        0           92s
antivaghash@Antivas-MBP addressbook-demo % kubectl apply -f Deployment.yaml
deployment.apps/addressbook configured
antivaghash@Antivas-MBP addressbook-demo % kubectl get pods
NAME                READY   STATUS    RESTARTS   AGE
addressbook-5686d576f9-zhm5  1/1     Running        0          3658s
antivaghash@Antivas-MBP addressbook-demo % kubectl expose deployment addressbook --type=NodePort --port=8080
service/addressbook exposed
antivaghash@Antivas-MBP addressbook-demo % kubectl get services
NAME                TYPE                CLUSTER-IP      EXTERNAL-IP      PORT(S)          AGE
addressbook         NodePort            10.109.289.9     <none>            8080:32285/TCP   11s
kubernetes           ClusterIP           10.96.0.1        <none>            443/TCP          9m54s
antivaghash@Antivas-MBP addressbook-demo %

```

kubectl port-forward services/addressbook 7080:8080

```

> preloaded-images-k8s-v18-v1...: 341.16 MiB / 341.16 MiB 100.00% 61.07 M
Creating docker container (CPUs=2, Memory=4096MB) ...
Preparing Kubernetes v1.28.3 on Docker 24.0.7 ...
  • Generating certificates and keys ...
  • Booting up control plane ...
  • Configuring RBAC rules ...
  • Configuring bridge CNI (Container Networking Interface) ...
  • Verifying Kubernetes components...
  • Using image gcr.io/k8s-minikube/storage-provisioner:v5
  Enabled add-ons: storage-provisioner, default-storageclass
Kubectl not found. If you need it, try: 'minikube kubectl -- get pods -A'
Done! Kubectl is now configured to use "minikube" cluster and "default" namespace by default
antivaghash@Antivas-MBP addressbook-demo % kubectl get services
antivaghash@Antivas-MBP addressbook-demo % kubectl get pods
> kubectl get pods
> kubectl: 64.8 / 64.8 MiB [-----] 100.00% 0.01 MiB p/s 11s
NAME                TYPE                CLUSTER-IP      EXTERNAL-IP      PORT(S)          AGE
kubernetes           ClusterIP           10.96.0.1        <none>            443/TCP          57s
antivaghash@Antivas-MBP addressbook-demo % kubectl get pods -A
NAMESPACE   NAME                 READY   STATUS    RESTARTS   AGE
kube-system  coredns-5dd575ab8-c4pgn  1/1     Running   0           71s
kube-system  etcd-minikube         1/1     Running   0           88s
kube-system  kube-apiserver-minikube  1/1     Running   0           88s
kube-system  kube-controller-manager-minikube  1/1     Running   0           88s
kube-system  kube-proxy-rzk22      1/1     Running   0           71s
kube-system  kube-scheduler-minikube  1/1     Running   0           88s
kube-system  storage-provisioner    1/1     Running   1 (67s ago)  84s
antivaghash@Antivas-MBP addressbook-demo % vi Deployment.yaml
antivaghash@Antivas-MBP addressbook-demo % kubectl create -f Deployment.yaml
deployment.apps/addressbook created
antivaghash@Antivas-MBP addressbook-demo % kubectl get pods
NAME                READY   STATUS    RESTARTS   AGE
addressbook-5686d576f9-zhm5  0/1     ContainerCreating   0           6s
antivaghash@Antivas-MBP addressbook-demo % kubectl get pods
NAME                READY   STATUS    RESTARTS   AGE
addressbook-5686d576f9-zhm5  1/1     Running        0          12s
antivaghash@Antivas-MBP addressbook-demo % vi Deployment.yaml
antivaghash@Antivas-MBP addressbook-demo % kubectl apply -f Deployment.yaml
Warning: resource Deployment/addressbook is missing the kubectrl.kubernetes.io/last-applied-configuration annotation which is required by kubectl apply. kubectl apply should only be used on resources created directly by either kubectl create --save-config or kubectl apply. The missing annotation will be patched automatically.
deployment.apps/addressbook configured
antivaghash@Antivas-MBP addressbook-demo % kubectl get pods
NAME                READY   STATUS    RESTARTS   AGE
addressbook-5686d576f9-j854k  1/1     Running        0           6s
addressbook-5686d576f9-zhm5  1/1     Running        0          2022s
addressbook-5686d576f9-xgq2  1/1     Running        0           6s
antivaghash@Antivas-MBP addressbook-demo % vi Deployment.yaml
antivaghash@Antivas-MBP addressbook-demo % kubectl get pods
NAME                READY   STATUS    RESTARTS   AGE
addressbook-5686d576f9-j854k  1/1     Running        0           92s
addressbook-5686d576f9-zhm5  1/1     Running        0          3644s
addressbook-5686d576f9-xgq2  1/1     Running        0           92s
antivaghash@Antivas-MBP addressbook-demo % kubectl apply -f Deployment.yaml
deployment.apps/addressbook configured
antivaghash@Antivas-MBP addressbook-demo % kubectl get pods
NAME                READY   STATUS    RESTARTS   AGE
addressbook-5686d576f9-zhm5  1/1     Running        0          3658s
antivaghash@Antivas-MBP addressbook-demo % kubectl expose deployment addressbook --type=NodePort --port=8080
service/addressbook exposed
antivaghash@Antivas-MBP addressbook-demo % kubectl get services
NAME                TYPE                CLUSTER-IP      EXTERNAL-IP      PORT(S)          AGE
addressbook         NodePort            10.109.289.9     <none>            8080:32285/TCP   11s
kubernetes           ClusterIP           10.96.0.1        <none>            443/TCP          9m54s
antivaghash@Antivas-MBP addressbook-demo % kubectl port-forward services/addressbook 7080:8080
Forwarding from 127.0.0.1:7080 -> 8080
Forwarding from [::]:7080 -> 8080

```

8. Verify your application.

Run the application on localhost with the port being forwarded I.e. 7080 in this case.

